SuperVision - The world cinema explorer

Process Book

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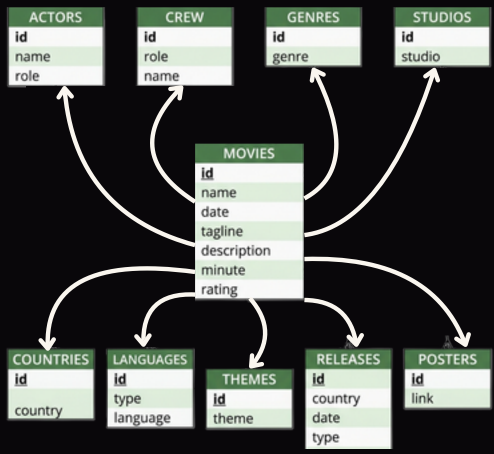
1. The dataset



We chose the Letterboxd Movies Dataset from Kaggle because all of use shared a deep interest in cinema and thought that the social dynamics behind Letterboxd had a lot to give to us. With **941,597 films** organized in a star schema.

The dataset was large enough to be statistically meaningful, multi-relational enough to support very different angles of analysis (geographic, temporal, social, visual) and complete enough on the fields that mattered most. That combination of scale, structure, and quality made it a perfect choice.

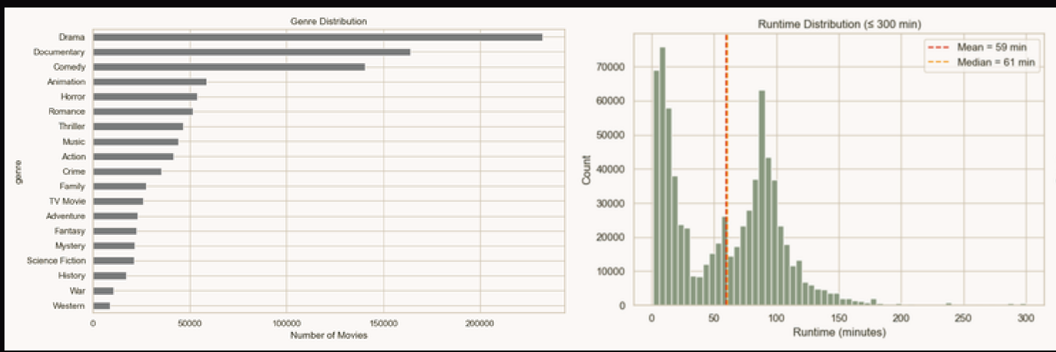
1.1 - Our exploratory data analysis

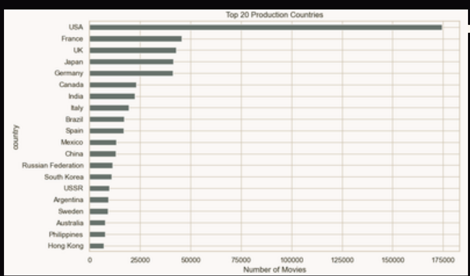
Our EDA surfaced several facts that would directly shape our design decisions:

- **Data quality & scope:** The dataset is broad but uneven in depth. The most important discovery was that 90% of movies have no average rating, because the Letterboxd community's rating history only extends back reliably to the late 1990s.

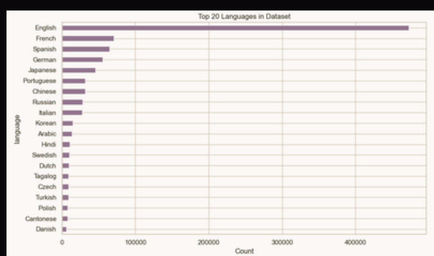
- **Short films are 37% of the corpus.** The runtime distribution has a sharp bimodal split, with a huge mass of films under 40 minutes alongside the expected feature-film peak around 90–100 minutes. Extreme outliers (entries of 72,000 minutes (50 days) were clearly data errors and required filtering.

- **Drama dominates,** but the landscape is rich. Drama accounts for 232K genre tags, followed by Documentary (164K) and Comedy (141K). Because each movie can carry multiple genre tags, the real story is in the combinations and how they shift over time and across geographies.





- The USA accounts for 174K movies which is 3.8× more than second-place France (46K). Yet among the top 15 producing countries, the USSR and Japan rate highest on average, almost certainly due to survivorship bias: only the great works of those countries were cataloged at all.



- **English is dominant but not total.** 45% of all language assignments are English. However, the spoken languages and main language fields paint a more nuanced picture that turned out to be one of the most interesting statistics in the country-level detail panels.

2. Evolution

In the **first milestone** we articulated three analytical lenses we wanted to combine into a single interactive experience:

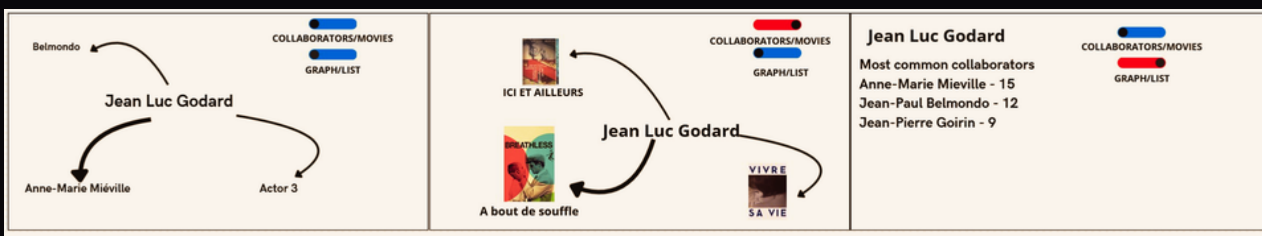
- **Geographic**: we wanted to bring out the differences and specialities of local cinema across the world (highest rated films, most prolific directors, most popular genres, ...) but also allow people to discover movies and directors from smaller countries.
- **Temporal**: how cinema has evolved over time: movie length, genres distribution, cast and crew size, etc.
- **Social**: the collaborative networks that define artistic movements, identifying filmmakers who cluster together and the films that connect them.

We added to this a fourth lens, **Visual**, was added later in the form of The Palette (dominant colors extracted from movie posters).

Our **second milestone** outlined three core visualizations:

1. **Interactive World Map:** Countries encoded by color saturation (average film rating or total output). Clicking a country opens a linked detail panel: highest-rated films, top actors/directors, genre distribution. The early sketch showed a flat mercator projection with a side panel.

2. **Collaboration Graph:** A force-directed graph centered on a chosen filmmaker. Edges radiating to all collaborators, with thickness+opacity proportional to shared film count. We planned two toggle modes: "Graph / List" and "Collaborators / Movies".



3. **Movie Detail Panel:** A shared side panel (reachable from both the map and the network) showing a single film entry: poster, director, release year, cast, and rating. The semantic bridge between geographic and relational perspectives, illustrated in the sketch with: Ici et ailleurs (Godard, 1976).

To go further -

This second milestone report also described several planned extensions:

- **Time Slider** (range 1874–2025) to filter all views by year range
- **Co-production Flow Lines:** directional arcs between countries weighted by joint film count
- **Period Visual Effects:** CSS filters that shift with the era (grayscale for pre-1950, saturated for 1950s Technicolor, grain for the 1970s)
- **Visual Poster Wall:** clusters of visually similar film posters sorted by extracted color

Implementation Choices at M2:

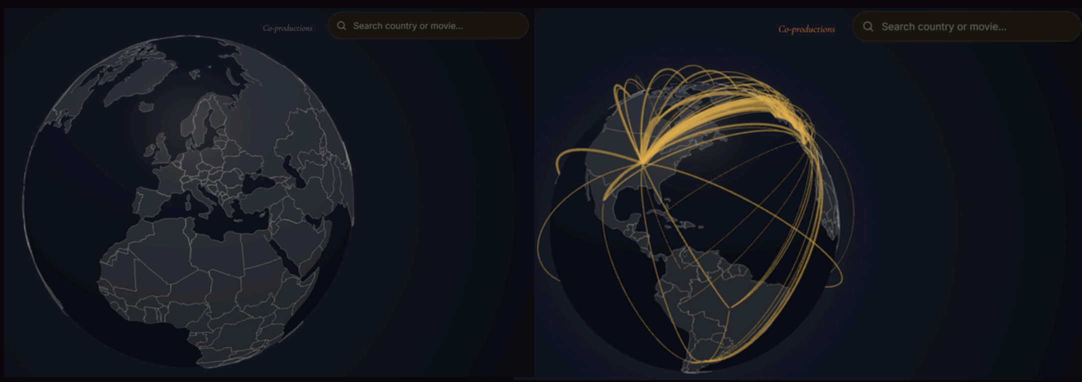
At Milestone 2 we committed to:

- globe.gl (wrapping Three.js) for the 3D globe
- D3-force for the network simulation
- Nuxt 4 (.vue) as the application framework
- A SQLite backend queried via Nuxt server routes (avoiding the need to ship a multi-GB JSON blob to the client)

3. From Plan to Product

3.1 The Globe

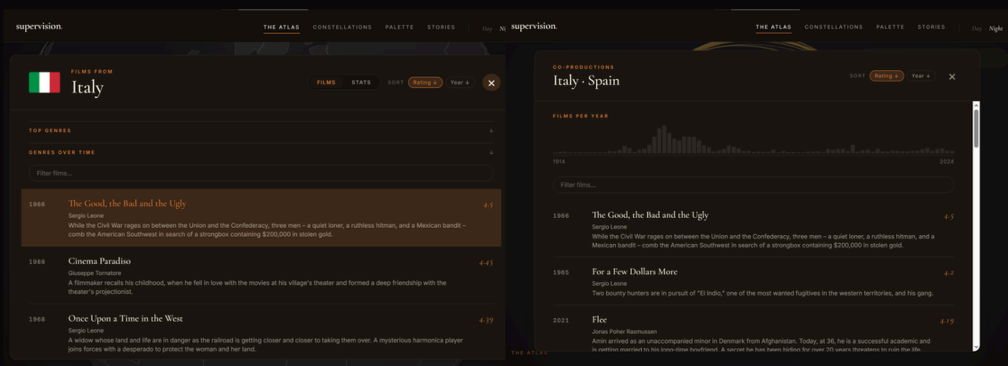
Our original plan was to build a flat choropleth map using D3's geographic projections, colored by a single metric such as average rating or film count. However, we shifted from this approach to create a fully interactive 3D globe using `globe.gl`. Instead of encoding a specific data variable across countries with a traditional choropleth coloring, which we realized tended to wash out into an unreadable sea of uniform brown across 200 countries, we prioritized interactive exploration. The final implementation features a toggleable overlay of co-production arcs connecting country pairs (with stroke width mapped to the square root of co-production volume) and a global search bar that navigates directly to countries or films. Clicking any country triggers a smooth zoom transition to that region, opening a panel detailing its cinema industry. While clicking on any link in co-productions mode allows us to see the detail about the collaborations between those two country.



3.2 The Country Panel

To support this exploratory workflow, we developed two specialized detail interfaces:

- a country-specific panel and a bilateral co-production modal. The country panel slides in as a two-tabbed dashboard, dividing the data between browsing and analysis. Under the "Films" tab, users can filter the country's catalog using an interactive genre donut chart or a stacked bar chart showing genre proportions over time, linking directly to a paginated, searchable movie list. The "Stats" tab acts as an analytical workspace, displaying key metrics (average rating, runtime, co-production rates), distribution metrics (languages, themes, partners), and custom-engineered SVG line charts tracking historical trends in crew sizes, ratings, and genre popularity.
- Clicking a co-production arc reveals the specific collaborative footprint between two nations. It features a temporal histogram showing films co-produced per year, which acts as an interactive year-filter, and a sortable list of the joint productions, allowing users to drill down into film descriptions, directors, and ratings. Together, these detail panels transform spatial exploration on the globe.

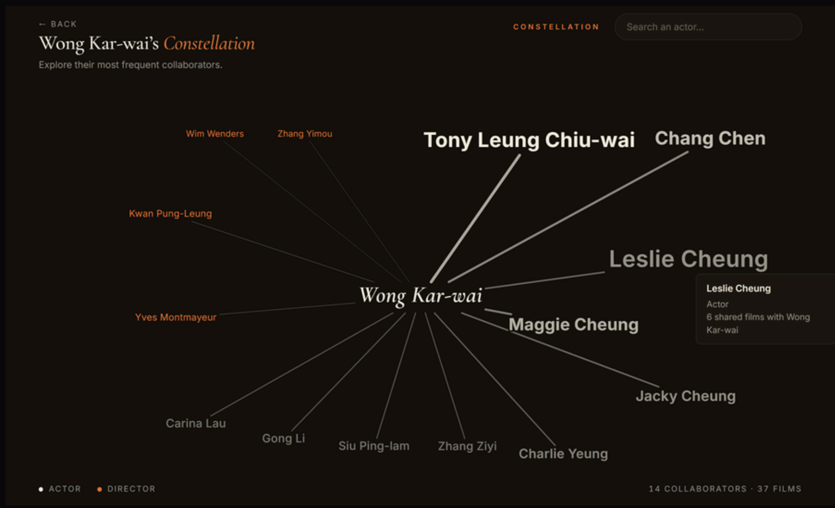


3.3 The Collaboration Graph

Our original plan for the collaboration network was a single, open-search graph where any filmmaker's name could be typed to display their professional network, toggleable between graph and list views. The final implementation, named "Constellations," maintains this open-search capability, allowing users to search for any actor or director in the dataset. However, visualizing a highly prolific filmmaker's entire network (such as actors with over a thousand film credits) would result in a dense, illegible "spaghetti" graph of hundreds of overlapping edges.

To solve this visual complexity and performance issue, we implemented a server-side programmatic limit rather than restricting search: the SQLite backend dynamically fetches and constrains the network to the center filmmaker's top 10 co-actors and top 4 co-directors. This hybrid approach pre-computes static JSON files for key featured actors to enable instant loading, while querying the live database for any other filmmaker to generate a clean, maximum 15-node constellation. Nodes are styled based on collaboration frequency, with directors accented in burnt orange and actors styled in neutral ink.

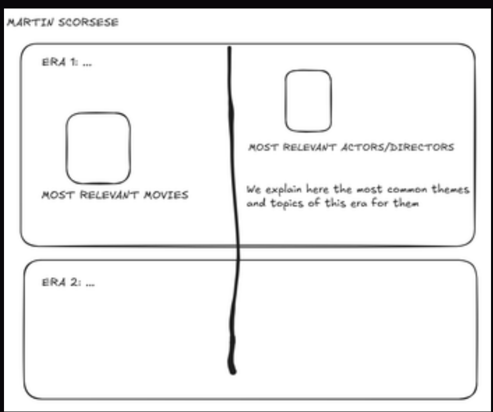
The graph integrates advanced interactions: hover triggers a fisheye distortion that scales adjacent nodes up to 1.7× for easier reading, nodes can be dragged and pinned, and clicking a node slides in a sidebar displaying shared films as a poster grid. Crucially, the simulation is sidebar-aware, recalculating and shifting its center point dynamically to prevent nodes from being occluded when the drawer opens, while rendering is managed through CSS custom properties to adapt instantly to theme changes.



3.4 The Storylines View

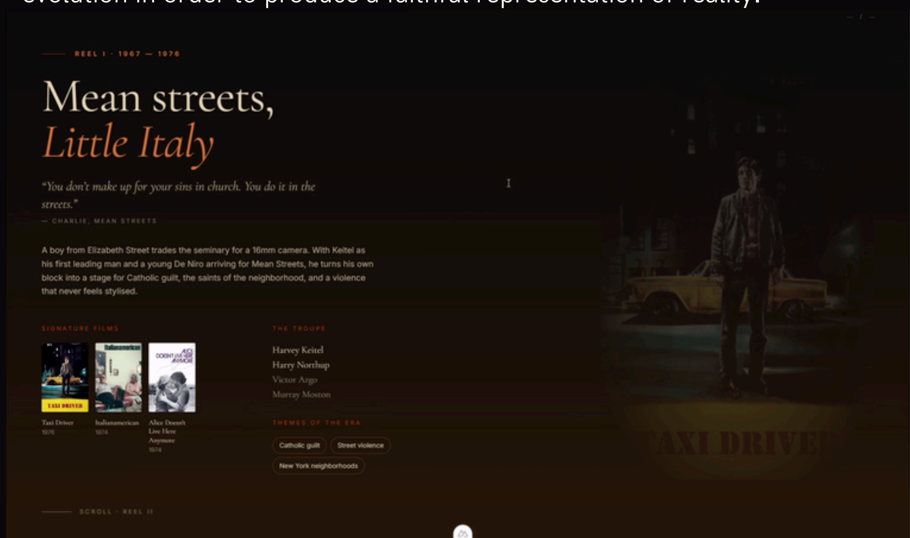
The "Storylines" section was added to weave raw data into curated, chronological narratives. We built full-viewport, scroll-snapping chapters for six featured filmmakers, transitioning from a typographic cover through creative era panels featuring horizontal poster reels, and ending with a coda linking back to their geographic and network views.

Each storyline was designed to guide viewers through the life and artistic evolution of influential cinematic figures such as Martin Scorsese, creating an immersive experience that allowed them to quickly grasp the key personalities, themes, and films that defined each era.



Initial Storyline Sketch

We worked on a total of six storylines, aiming to capture the essence of each major figure in cinema within these short narratives. The process required briefly researching each creator's biography and artistic evolution in order to produce a faithful representation of reality.



Resulting product

3.5 The Palette

The "Visual Poster Wall" from the Milestone 2 plan was the hardest feature to implement and the last to be implemented. The original idea would have required running a dimension-reduction algorithm (UMAP or t-SNE) on extracted image embeddings, which was beyond our infrastructure.
What we shipped instead was **The Palette**: a single canvas element rendering one horizontal stripe per year (1920–2024), where each stripe shows the dominant colors extracted from that year's movie posters, sorted left to right by HSL hue. Grays are pushed to the right end. The result is a continuous visual timeline of cinema's color history, you can trace the shift from the sepia/monochrome palette of the 1920s–40s, through the vivid saturations of the 1950s–60s, into the desaturated earth tones of the 1970s, and the increasingly digital blues and teals of the 2000s.

4. Visual and Interactive Design

Our aesthetic approach was conceived to bridge the gap between statistical analysis and cinematic editorial layout. Rather than adopting a clinical, corporate dashboard style, we structured the platform to feel like a film journal.
We established a visual system built around a clear environmental contrast: a warm parchment background for general exploration, evoking physical print media and film critiques, which shifts to a theater-like near-black for narrative chapters to simulate a projection room. These environments are tied together by a desaturated burnt-orange accent that references both the warmth of a projector lamp and Letterboxd's green-and-orange brand identity. Typography contrasts a classic, high-contrast literary serif for editorial titles and subtitles with a clean, functional sans-serif for labels, interactive charts, and search fields.

This visual framework is reinforced by three interactive principles: screen focus, data on demand, and progressive disclosure.

- First, we restricted each view to a single primary action (spinning the globe, dragging network nodes, scrolling the narrative, or panning the color timeline) preventing the cognitive overload of competing layouts.
- Second, because our Letterboxd dataset contains nearly a million films, shipping the raw tables to the client was impossible; we implemented a strict "data on demand" paradigm where Vue components trigger lightweight SQLite queries only on explicit clicks (such as opening a country panel or loading a collaborator's filmography).
- Lastly, we structured information through progressive disclosure, ensuring that dense statistical layouts and list tables remain hidden behind clean entry points until the user initiates a deeper inspection.

5. Technical Challenges

5.1 The Globe's WebGL Display

globe.gl runs a Three.js WebGL renderer inside a persistent <canvas>. When the user navigates away from the Atlas page and back, Vue's component lifecycle unmounts and remounts the WorldGlobe component — but the WebGL context and GPU memory are not automatically cleaned up. We encountered a "too many WebGL contexts" error in development that caused the globe to go blank on the second visit.
The fix required a detailed onUnmounted hook that manually traverses the Three.js scene graph, calls .dispose() on every geometry, material, and texture, calls forceContextLoss() on the renderer, and physically removes the canvas DOM element before Vue's reconciler has a chance to reuse it.

5.2 Audio display following the Stories

Adding ambient audio to the storylines introduced several constraints beyond simply playing a sound file: modern browsers block autoplay unless it is triggered by a direct user interaction, so scrolling alone cannot reliably start audio, which we handled by requiring an opt-in toggle and providing a fallback that silently disables playback if permission is denied.

We also had to map continuous scroll position to a single "active" track using intersection-based detection of each section while remaining stable during fast scrolling; switching between tracks needed to feel smooth, so we implemented crossfading using gradual volume changes across overlapping looping audio elements, making sure to cancel any conflicting transitions

6. Peer Assesment

Team member	**Contribution**
Néhémie Frei	Worked on the stats view of every movie, proposed most of the initial ideas and prototypes, and worked on the storylines
Clara Hamousz	Worked on the co-productions, cinema palette and movie pannel and chose the visual theme of the website
Arthur Taieb	Designed and implemented the Atlas globe interface, worked on the movie pannel, and designed the process book, also recorded the screen cast
Adrian Martinez	Mainly worked on building the actor graph view, creating some of the storylines, implementing the DB integration and optimizing the website's performance

Note: Contributions overlapped substantially during integration, the division above reflects primary ownership, not exclusive authorship.

7. Conclusion

The final product is meaningfully different from what we planned in Milestone 1. The flat, static choropleth became a three-dimensional spinning globe. The open filmmaker search was enriched with dynamic network-restriction algorithms to keep graphs readable, and paired with an entirely new editorial chapter section. The planned poster wall crystallized into a continuous visual timeline capturing a century of color trends in cinema. Even the planned global time slider was replaced by a richer, more context-aware set of historical trend charts integrated directly into individual country panels. These changes were not problems; they were responses to the shape and limitations of the data itself.

What stayed constant throughout these design cycles was our core aspiration: to make world cinema discoverable, not just measurable. The **Atlas** lets you spin the globe and land on a country you know nothing about, immediately uncovering its finest films, genre proportions, and international co-productions. The **Stories** offer an editorial doorway into the careers of iconic directors and actors. The **Constellations** map out the hidden social fabric of the industry, revealing how networks of collaborators shape artistic movements. Finally, the **Palette** provides a wordless visual history of movie poster aesthetics across a hundred years.

We are proud of the balance we found between visual storytelling and raw database analysis, delivering an interactive experience that honors both the volume of the dataset and the word cinema.